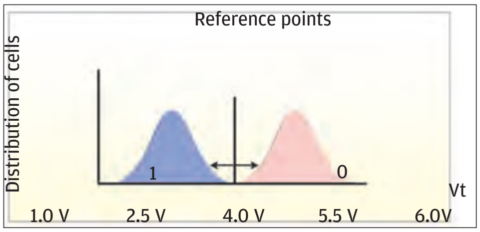
devices today: Single-Level Cells (SLC) and Multi-Level Cells (MLC).

Single-Level Cells

An SLC flash is so designed as to store only one bit-value per cell. So, for example, if the memory is being read, each cell can provide either of two values, depending on the voltage level – 0 (zero) if the cell is programmed or 1 (one) if the cell is blank or erased. Thus, the cell sensor in SLCs has only two voltage levels to bother about.



The SLC Flash detects a zero or a one based on the Threshold Voltage of the cell

Multi-Level Cells

Manufacturers, like Intel, use the same transistors for their SLC and MLC flash. The difference is in how data is stored or read. In the case of a Multi-Level Cell, every cell can represent four states:

a) 00 (two zeroes) indicate that the cell is fully programmed

b) 01 (a zero and a one) represent a partially programmed cell

c) 10 (a one and a zero) show that the cell is partly erased

d) 11(two ones) indicate that the cell is fully erased

In short, an MLC can store two bits of information with no difference in requirements of space or power. While the minimum and maximum voltage stay the same in SLCs and MLCs, there are more gradations between the two voltage levels, making the drive sensitive to smaller variations in voltage, allowing it to register two bits per cell. This, of course, would entail better programming algorithms, exact placement of charge and better charge sensing to detect the amount of charge on the floating gate more precisely. In an MLC, it is even possible to store more bits than two in a single cell. As the sensitivity to voltage deltas gets better, the flash drive simply needs to be able to detect 2N states in the cell (where N = the number of desired per cell). So, for example, if you want each cell to hold 4 bits, you'll need to be able to identify 24 (or sixteen) voltage states in the cell – viz. 0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111.

SLCs vs. MLCs

Let's take a better look at the differences between SLC and MLC flash drives.

MLCs take longer time to read and write data because it needs to track more minute gradations. SLCs can read and write data much faster. However, an MLC flash can hold double the number of transistors on the same wafer size as compared to an SLC flash. This allows greater density and thus, more memory on the same size chip bringing down the cost per MB. Cheap thumb drives or pen drives, therefore, prefer MLC.

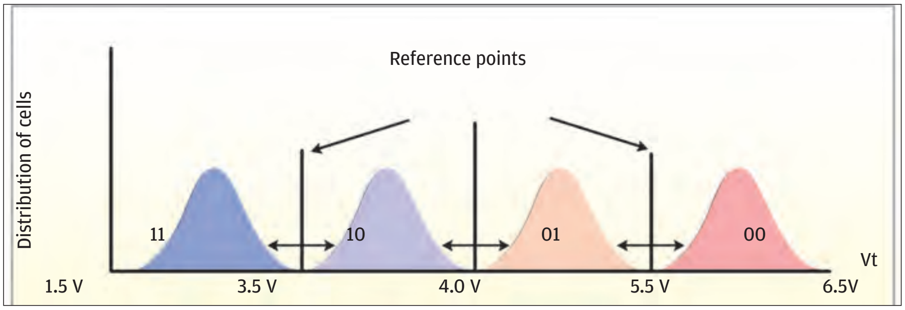
SLCs, however, score on one parameter – lifespan. Every flash drive has a limited number of erase/ programme cycles. This is because the silicon oxide material that is used to separate the Floating Gate from other parts of the cell suffers wear and tear with the repeated 'tunnelling' of electrons during erasure or programming. By and by, the material dissipates and electrons don't move about the way they should. Note that reading does not reduce life of the flash memory because the process does not involve the passage of electrons through the oxide, only detection of voltage difference.

All SSDs erase in blocks (groups of pages) and write in pages (groups of cells). At the rate we go about juggling data in our flash drives, the average drive should wear out really fast. Luckily, SLC flash drives deteriorate only after about 100,000 write/ erase cycles – which could last decades, unless you insist on rewriting all day and night for several years at a stretch (MLCs last for only about 10,000 such cycles). Deleting data is different. When you erase data, the drive doesn't really erase it, but only marks a set of pages as 'invalid' and stops showing them up when you open the drive folder on your screen. It then waits until you write in some data, removes the data from the 'erased' cells and puts the new bit values into them. This reduces the wear and tear and increases the lifespan of the drive.

Of course, not all drives delete the same way. Some poorly built flash drives have high Write Amplification factors – which is a technical way of talking about how much 'writing' actually happens. Therefore, if you want to change just a few pages of data, the drive will erase the entire block that contains those pages, make the changes and write the new block in, thus increasing the number of times 'tunnelling' happens. This reduces the lifespan of the drive.

To put it briefly, MLCs have a high density and can hold more data at cheaper costs and smaller sizes, while SLCs need less power, are more durable and allow faster writing and erasing of data. Choice a particular kind of drive would depend on your requirements.

Flash memory is not without its drawbacks. Some cells get unintentionally programmed while programming other adjacent ones (a phenomenon called 'Program Disturb'). Sometimes the threshold value is misread and a '1' is recorded instead of a '0' (called 'Read Disturb'). Charge may leak out of the FG, affecting data retention. Some question the ability of flash drives to store information when it comes to less than 30 nanometres. Manufacturers all over the world are working to better the existing technology and eliminate the drawbacks of both, MLCs as well as SLCs. For all practical purposes, NAND will do for the average consumer for the next few years at least. Flash is the future. d



MLCs give you two bits for the price of one